Revision Notes

CBSE Class 12 Physics • NCERT Part II, Chapters 9–14

Ray Optics • Wave Optics • Dual Nature • Atoms • Nuclei • Semiconductors

How to use these notes. Keep the separate **Formula Sheet** next to you. Each chapter has:

- **A Derivations**: short step outlines of the derivations the board asks (Ch 14: key explanations). Write them out once from these steps.
- **B Diagrams**: the figures to practise, with the labels that carry marks.
- **C Numerical patterns**: every NCERT exercise grouped into patterns, each with one solved question from the book. The small line under the boxes lists which questions belong to which pattern.
- **D Conceptual questions**: "why" questions with one-line answers. **E Exam tips**. **F JEE extras** (purple; not needed for the board exam).
- **G Pattern spotter**: a one-look table for final revision.

Question numbers match the NCERT textbook (Reprint 2026–27).

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Chapter 9 Ray Optics and Optical Instruments

9A Derivations (board favourites)

1. Mirror equation

Object AB, image A′B′ by a concave mirror; similar triangles A′B′F, MPF and A′B′P, ABP (paraxial rays, MP ≈ AB):

$\frac{\mathrm{B'A'}}{\mathrm{BA}} = \frac{\mathrm{B'F}}{\mathrm{FP}} = \frac{\mathrm{B'P}}{\mathrm{BP}} \Rightarrow \frac{v-f}{f} = \frac{v}{u}$ (with signs) $\Rightarrow \boxed{\frac{1}{v}+\frac{1}{u}=\frac{1}{f}}$, $m = -\frac{v}{u}$. Also $f = \frac{R}{2}$.

2. Refraction at a spherical surface

Small angles: $i = \alpha + \gamma$, $r = \gamma - \beta$; Snell $n_1 i = n_2 r$; with $\alpha \approx \frac{\mathrm{MN}}{-u}$, $\beta \approx \frac{\mathrm{MN}}{v}$, $\gamma \approx \frac{\mathrm{MN}}{R}$:

$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$.

3. Lens maker's formula and thin lens formula

Apply result 2 at the first surface (image at v_1), then at the second (object at v_1), add:

$\frac{1}{v} - \frac{1}{u} = (n_{21}-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. For $u = \infty$, $v = f$: $\boxed{\frac{1}{f} = (n_{21}-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$, so $\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}}$.

4. Lenses in contact

Image by lens 1 at v_1 is the object for lens 2: $\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$, $\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$. Add: $\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$,

$P = P_1 + P_2$, $m = m_1 m_2$.

5. Prism formula

$r_1 + r_2 = A$; deviation $\delta = (i - r_1) + (e - r_2) = i + e - A$. At minimum deviation $i = e$, $r_1 = r_2 = \frac{A}{2}$, $i = \frac{A + D_m}{2}$:

$\boxed{n = \frac{\sin\frac{A+D_m}{2}}{\sin\frac{A}{2}}}$. Thin prism: $\delta = (n-1)A$.

6. Magnifying power of microscope and telescope

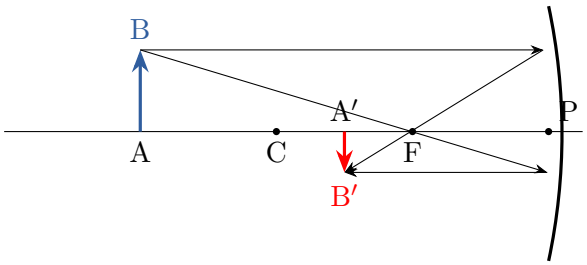
Simple microscope: $m = 1 + \frac{D}{f}$ (image at D); $\frac{D}{f}$ (image at ∞).

Compound microscope: $m = m_o m_e = \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$ (at D); $\frac{L}{f_o}\cdot\frac{D}{f_e}$ (normal adjustment), $L \approx$ tube length.

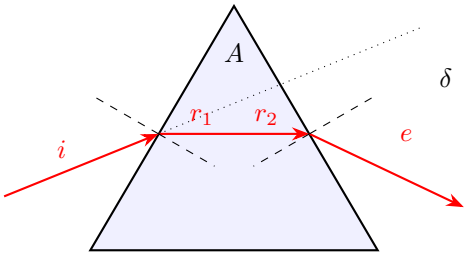
Astronomical telescope: normal adjustment $m = \frac{f_o}{f_e}$, $L = f_o + f_e$; image at D: $m = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$.

9B Diagrams to practise

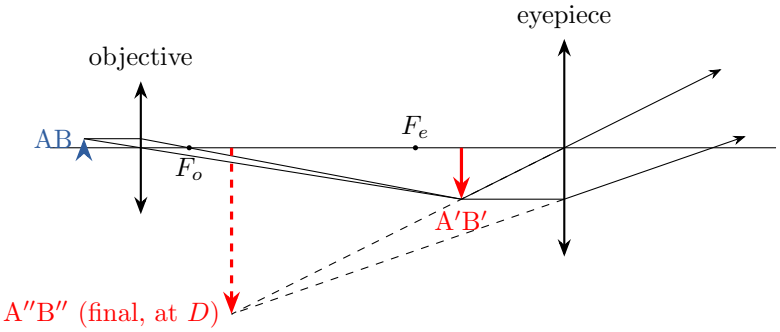
Concave mirror: object beyond C (labels: P, F, C, object AB, image A′B′, real and inverted)



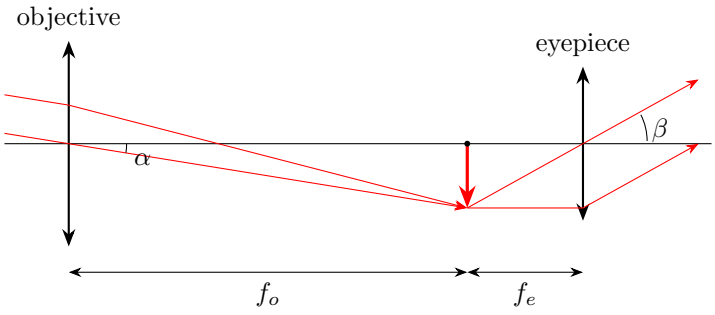
Refraction through a prism (labels: A, i, r_1, r_2, e, δ, normals)



Compound microscope (labels: objective, eyepiece, f_o, f_e, intermediate image A′B′, final image at D)



Astronomical telescope, normal adjustment (labels: f_o, f_e, common focus, α, β, final image at infinity)



Also practise: convex lens image for an object between F and 2F; total internal reflection in an optical fibre; the Cassegrain reflecting telescope.

9C Exercise-wise numerical patterns

Exercises 9.1–9.31 (31 questions, 6 patterns)

Pattern A: Mirror formula (and rotating mirrors)

Method: New Cartesian signs: distances against incident light are negative; concave $f<0$, convex $f>0$.

Solved (Q9.1): $f=-18$ cm, $u=-27$ cm: $\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=-\frac{1}{18}+\frac{1}{27}=-\frac{1}{54}\Rightarrow v=-54$ cm. $m=-\frac{v}{u}=-2$: image 5.0 cm, real, inverted, magnified; screen 54 cm in front of the mirror. Moving the candle closer moves the screen farther away.

Pattern B: Refraction, apparent depth, total internal reflection

Solved (Q9.5): critical angle $\sin i_c=\frac{1}{1.33}\Rightarrow i_c=48.8^\circ$. Light escapes inside a circle of radius $r=h\tan i_c=0.80\times 1.14=0.91$ m.
Area $=\pi r^2=2.6$ m^2.
(Q9.16): shift by a slab $=t\left(1-\frac{1}{n}\right)=15\left(1-\frac{1}{1.5}\right)=5.0$ cm, independent of where the slab is.

Pattern C: Prism

Solved (Q9.6): $n=\dfrac{\sin\frac{60^\circ+40^\circ}{2}}{\sin 30^\circ}=\dfrac{\sin 50^\circ}{0.5}=1.53$.
In water: $n_{rel}=\frac{1.53}{1.33}=1.15$; $\sin\frac{60^\circ+D_m}{2}=1.15\times 0.5=0.576\Rightarrow D_m\approx 10^\circ$.

Pattern D: Lens formula, lens maker's formula, combinations

Solved (Q9.7): equiconvex: $\frac{1}{f}=(n-1)\frac{2}{R}\Rightarrow R=2(n-1)f=2\times 0.55\times 20=22$ cm.

(Q9.19, displacement method): $f=\dfrac{D^2-d^2}{4D}=\dfrac{90^2-20^2}{4\times 90}=21.4$ cm.

Pattern E: Magnifier and compound microscope

Solved (Q9.11(a)): eyepiece: $v_e=-25$, $f_e=6.25$: $\frac{1}{u_e}=\frac{1}{v_e}-\frac{1}{f_e}=-\frac{1}{25}-\frac{4}{25}\Rightarrow u_e=-5$ cm.
Objective image at $v_o=15-5=10$ cm; $\frac{1}{u_o}=\frac{1}{10}-\frac{1}{2}\Rightarrow u_o=-2.5$ cm.
$m=\dfrac{v_o}{|u_o|}\left(1+\dfrac{25}{f_e}\right)=4\times 5=20$. (b) Final image at ∞: $u_e=-6.25$, $v_o=8.75$, $u_o=-2.59$ cm, $m=13.5$.

Pattern F: Telescope

Solved (Q9.27, 9.28): (a) normal adjustment $m=\frac{f_o}{f_e}=\frac{140}{5}=28$; (b) image at 25 cm: $m=28\left(1+\frac{5}{25}\right)=33.6$.
Separation $=f_o+f_e=145$ cm. Tower subtends $\frac{100}{3000}=\frac{1}{30}$ rad; image by objective $h=f_o\times\frac{1}{30}=4.7$ cm.

Pattern A: Q9.1, 9.2, 9.15, 9.29, 9.30. Pattern B: Q9.3, 9.4, 9.5, 9.16, 9.17. Pattern C: Q9.6, 9.21. Pattern D: Q9.7, 9.8, 9.9, 9.10, 9.18, 9.19, 9.20, 9.31. Pattern E: Q9.11, 9.12, 9.22–9.26. Pattern F: Q9.13, 9.14, 9.27, 9.28.

9D Conceptual questions

- **Why do diamonds sparkle?** → Small critical angle ($\approx 24°$); cut so light undergoes repeated TIR.
- **Why is the sky blue / sun red at sunset?** → Scattering $\propto 1/\lambda^4$: blue scattered most; at sunset light travels farther, blue is removed.
- **Advantage of a reflecting telescope?** → No chromatic aberration, easier to support a large mirror, more light, better resolution.
- **Why should the objective of a telescope have a large aperture?** → More light gathered and better resolving power.
- **Focal length of a lens in water?** → Increases (smaller $n_{21} - 1$).
- **Why does a convex mirror serve as a rear-view mirror?** → Always erect, diminished image with a wide field of view.
- **Does a magnifying glass change the angle subtended?** → It lets the object come closer than 25 cm, so the angle is larger.

9E Exam tips

Common mistakes that lose marks

- Sign convention: write u, v, f, R with signs *before* substituting.
- Mirror: $\frac{1}{v} + \frac{1}{u}$; lens: $\frac{1}{v} - \frac{1}{u}$.
- Ray diagrams: arrows on every ray, dotted lines for virtual rays, label F, C/2F and the image nature.
- Microscope image at D: use $1 + \frac{D}{f_e}$; at infinity: $\frac{D}{f_e}$. Use $D = 25$ cm.
- Power in dioptre needs f in metres.

Question types: 1-mark: TIR conditions, sign of f, power. 2–3 marks: mirror/lens numericals, prism n. 5 marks: derivations (mirror equation, lens maker's, prism) and ray diagrams of microscope/telescope with magnifying power.

9F JEE extras

Useful results

- Displacement method: $f = \dfrac{D^2 - d^2}{4D}$; real image needs $D \geq 4f$ (Q9.18: $f_{max} = 0.75$ m).
- Rotating a mirror by θ turns the reflected ray by 2θ (Q9.30).
- Silvered lens: $\dfrac{1}{F} = \dfrac{2}{f_l} + \dfrac{1}{f_m}$ (mirror focal length with sign).
- Two lenses separated by d: $\dfrac{1}{F} = \dfrac{1}{f_1} + \dfrac{1}{f_2} - \dfrac{d}{f_1 f_2}$.
- Numerical aperture of a fibre: $\sin i_{max} = \sqrt{n_1^2 - n_2^2}$.

Example (Q9.17(a)): $\sqrt{1.68^2 - 1.44^2} = 0.865 \Rightarrow i_{max} \approx 60°$. ✓

9G Pattern spotter

If you see…	Do this	See
Mirrors and refraction		
Image by a mirror	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, $m=-\frac{v}{u}$	Ex A
Apparent depth / shift by a slab	$\frac{\text{real}}{\text{apparent}}=n$; shift $t(1-\frac{1}{n})$	Ex B
Light escaping from under water	$\sin i_c=\frac{1}{n}$, $r=h\tan i_c$	Ex B
Optical fibre	TIR at core–cladding: $\sin i_c=\frac{n_2}{n_1}$	Ex B
Prism and lenses		
Prism n from D_m	$n=\frac{\sin((A+D_m)/2)}{\sin(A/2)}$	Ex C
Prism in a liquid	Use $n_{rel}=n_{glass}/n_{liquid}$	Ex C
Radius for a given f	Lens maker's formula	Ex D
Image by a lens; converging beam	$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ (virtual object $u>0$)	Ex D
Lenses in contact	$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$	Ex D
Two positions of lens, fixed screen	$f=\frac{D^2-d^2}{4D}$	Ex D
Instruments		
Simple magnifier	$1+\frac{D}{f}$ or $\frac{D}{f}$	Ex E
Compound microscope	Work back from the eyepiece; $m=m_om_e$	Ex E
Telescope	$\frac{f_o}{f_e}$; $L=f_o+f_e$	Ex F
Size of image by objective	$h=f_o\times$ (angle subtended)	Ex F

Chapter 10 Wave Optics

10A Derivations (board favourites)

1. Snell's law from Huygens' principle

Plane wavefront AB hits the surface at angle i. While the wavelet from B travels $\mathrm{BC} = v_1\tau$ in medium 1, the wavelet from A travels $\mathrm{AE} = v_2\tau$ in medium 2.

$\sin i = \dfrac{v_1\tau}{\mathrm{AC}}$, $\sin r = \dfrac{v_2\tau}{\mathrm{AC}}$ $\Rightarrow$ $\boxed{\dfrac{\sin i}{\sin r} = \dfrac{v_1}{v_2} = \dfrac{n_2}{n_1}}$. Frequency unchanged; $\lambda_2 = \lambda_1 \dfrac{v_2}{v_1}$.

Reflection: the same construction gives $i = r$.

2. Young's double slit: fringe positions and width

Path difference at P (distance x from centre): $S_2P - S_1P \approx \dfrac{xd}{D}$.

Bright: $\dfrac{xd}{D} = n\lambda \Rightarrow x_n = \dfrac{n\lambda D}{d}$. Dark: $\dfrac{xd}{D} = \left(n + \tfrac{1}{2}\right)\lambda$. Fringe width $\boxed{\beta = \dfrac{\lambda D}{d}}$.

3. Intensity in interference

Waves $a\cos\omega t$ and $a\cos(\omega t + \phi)$: resultant amplitude $2a\cos\frac{\phi}{2}$, so $\boxed{I = 4I_0\cos^2\dfrac{\phi}{2}}$, $\phi = \dfrac{2\pi}{\lambda} \times$ path difference.

Maxima $4I_0$ at $\phi = 2n\pi$; minima 0 at $\phi = (2n+1)\pi$. Incoherent sources: $I = 2I_0$ (no fringes).

4. Single-slit diffraction

Divide the slit of width a into two halves: at θ with $a\sin\theta = \lambda$ each point in one half cancels a point in the other: **first minimum** $\theta = \dfrac{\lambda}{a}$.

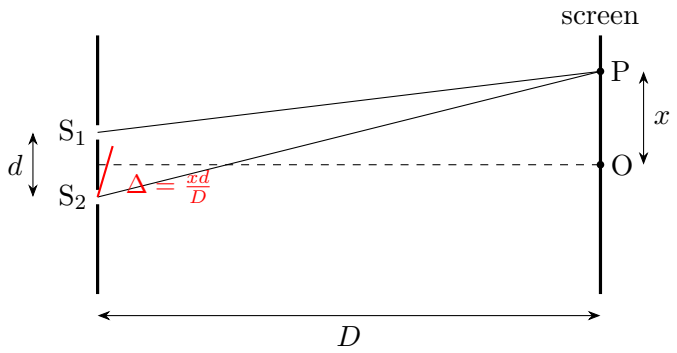
Minima: $\theta = \dfrac{n\lambda}{a}$; secondary maxima: $\theta \approx \left(n + \tfrac{1}{2}\right)\dfrac{\lambda}{a}$. Central maximum width $= \dfrac{2\lambda D}{a}$ (twice the others).

5. Polarisation (Malus' law)

Unpolarised light through a polaroid: intensity halves. Polarised light at angle θ to the pass-axis: $\boxed{I = I_0\cos^2\theta}$.

10B Diagrams to practise

Young's double-slit geometry (labels: S_1, S_2, d, D, x, P, O, path difference)



Also practise: Huygens construction for refraction of a plane wave (rarer to denser); intensity graph of single-slit diffraction (broad central maximum); polaroid arrangement.

10C Exercise-wise numerical patterns

Exercises 10.1–10.6 (6 questions, 3 patterns)

Pattern A: Light in a medium; wavefronts

Rule: frequency never changes on refraction; $v = \dfrac{c}{n}$, $\lambda = \dfrac{\lambda_0}{n}$.

Solved (Q10.1): reflected: $\lambda = 589$ nm, $\nu = \dfrac{3\times10^8}{589\times10^{-9}} = 5.09\times10^{14}$ Hz, $c = 3\times10^8$ m/s.
Refracted: $\nu = 5.09\times10^{14}$ Hz, $v = \frac{3\times10^8}{1.33} = 2.26\times10^8$ m/s, $\lambda = \frac{589}{1.33} = 444$ nm.

Pattern B: YDSE fringe positions

Solved (Q10.4): $x_4 = \dfrac{4\lambda D}{d} \Rightarrow \lambda = \dfrac{x_4 d}{4D} = \dfrac{1.2\times10^{-2}\times0.28\times10^{-3}}{4\times1.4} = 6.0\times10^{-7}$ m $= 600$ nm.
Coinciding fringes (Q10.6(b)): find the smallest $n_1\lambda_1 = n_2\lambda_2$: $n_1\times650 = n_2\times520 \Rightarrow n_1 = 4$, $n_2 = 5$.

Pattern C: Intensity and phase

Solved (Q10.5): path difference $\lambda \Rightarrow \phi = 2\pi$, $I = 4I_0\cos^2\pi = 4I_0 = K$.
Path difference $\frac{\lambda}{3} \Rightarrow \phi = \frac{2\pi}{3}$: $I = 4I_0\cos^2\frac{\pi}{3} = 4I_0\times\frac{1}{4} = \dfrac{K}{4}$.

Pattern A: Q10.1, 10.2, 10.3. Pattern B: Q10.4, 10.6. Pattern C: Q10.5.

10D Conceptual questions

- **Why are two independent bulbs not coherent?** → Their phase difference changes randomly and rapidly.
- **Effect on fringe width of: increasing d / D / λ; immersing in water?** → $\beta\downarrow$ / $\beta\uparrow$ / $\beta\uparrow$; β falls by factor n.
- **White light in YDSE?** → Central fringe white, others coloured (red outermost in each order).
- **Narrowing the slit in single-slit diffraction?** → Central maximum becomes wider.
- **Interference vs diffraction?** → Interference: two sources, equal-intensity fringes. Diffraction: one wavefront, intensity falls off from the central maximum.
- **Does energy get destroyed at dark fringes?** → No; it is redistributed to the bright fringes.
- **Which colour travels slower in glass?** → Violet ($n_v > n_r$).

10E Exam tips

Common mistakes that lose marks

- Frequency stays the same in a new medium; only v and λ change.
- "nth bright fringe from centre" is at $n\beta$; "nth dark fringe" is at $(n-\frac{1}{2})\beta$.
- Convert mm and nm to metres.
- Intensity uses $\cos^2(\phi/2)$, not $\cos^2\phi$.

Question types: 1-mark: wavefront shapes, coherence, Malus' law. 2–3 marks: YDSE numericals, intensity ratio, Huygens construction. 5 marks: Snell's law from Huygens, YDSE fringe width derivation, single-slit diffraction.

10F JEE extras

Useful results

- Unequal intensities: $\dfrac{I_{max}}{I_{min}} = \left(\dfrac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2$.
- Thin sheet (t, n) over one slit: fringe shift $= \dfrac{(n-1)tD}{d}$.
- Brewster's law: $\tan i_B = n$; reflected and refracted rays are perpendicular.
- Fresnel distance: $z_F = \dfrac{a^2}{\lambda}$.

Example: $I_1 = 4I_2$: $\frac{I_{max}}{I_{min}} = \left(\frac{2+1}{2-1}\right)^2 = 9$.

10G Pattern spotter

If you see…	Do this	See
Waves in media		
λ, ν, v in a medium	ν same; $v = c/n$; $\lambda = \lambda_0/n$	Ex A
Shape of a wavefront	Point source: spherical; at focus of lens / distant: plane	Ex A
Interference and diffraction		
Distance of nth bright fringe	$x_n = n\lambda D/d$	Ex B
Find λ from fringe data	$\lambda = xd/(nD)$	Ex B
Two wavelengths, fringes coincide	$n_1\lambda_1 = n_2\lambda_2$	Ex B
Intensity at a path difference	$I = I_{max}\cos^2(\phi/2)$	Ex C
Single slit minima	$a\sin\theta = n\lambda$	10A
Light through polaroids	$I = I_0\cos^2\theta$; unpolarised halves	10A

Chapter 11 Dual Nature of Radiation and Matter

11A Derivations (board favourites)

1. Einstein's photoelectric equation

A photon of energy $h\nu$ gives all its energy to one electron; ϕ_0 is used to escape:
$\boxed{K_{max} = h\nu - \phi_0}$, $K_{max} = eV_0 \Rightarrow V_0 = \dfrac{h}{e}\nu - \dfrac{\phi_0}{e}$ (straight line; slope h/e). Threshold: $\nu_0 = \dfrac{\phi_0}{h}$.

2. Photon properties

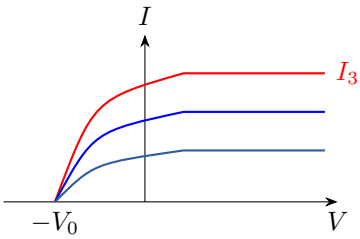
$E = h\nu = \dfrac{hc}{\lambda}$; $p = \dfrac{h\nu}{c} = \dfrac{h}{\lambda}$; rest mass zero; number per second $N = \dfrac{P}{h\nu}$.

3. de Broglie wavelength

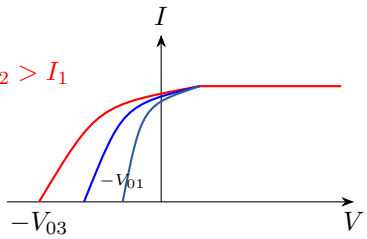
$\lambda = \dfrac{h}{p} = \dfrac{h}{mv}$. From kinetic energy: $\lambda = \dfrac{h}{\sqrt{2mK}}$. Electron accelerated through V: $\lambda = \dfrac{h}{\sqrt{2meV}} = \boxed{\dfrac{1.227}{\sqrt{V}}\ \text{nm}}$.

11B Diagrams to practise

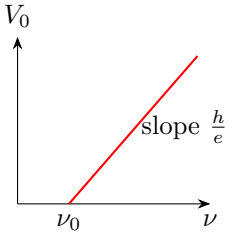
Photoelectric graphs (labels: saturation current, stopping potential $-V_0$, ν_0, slope h/e)



same ν, different intensities



same intensity, $\nu_3 > \nu_2 > \nu_1$



V_0 vs ν

11C Exercise-wise numerical patterns

Exercises 11.1–11.11 (11 questions, 3 patterns)

Pattern A: Photon energy, momentum and number

Solved (Q11.4): $E = \dfrac{hc}{\lambda} = \dfrac{6.63\times10^{-34}\times3\times10^{8}}{632.8\times10^{-9}} = 3.14\times10^{-19}$ J; $p = \dfrac{h}{\lambda} = 1.05\times10^{-27}$ kg m/s.

Photons per second $N = \dfrac{P}{E} = \dfrac{9.42\times10^{-3}}{3.14\times10^{-19}} = 3\times10^{16}$. H atom with same p: $v = \dfrac{p}{m_H} = \dfrac{1.05\times10^{-27}}{1.67\times10^{-27}} = 0.63$ m/s.

(Q11.1): $\nu_{max} = \dfrac{eV}{h} = 7.24\times10^{18}$ Hz; $\lambda_{min} = \dfrac{c}{\nu_{max}} = 0.041$ nm.

Pattern B: Einstein's equation (work function, threshold, stopping potential)

Solved (Q11.2): $h\nu = \dfrac{6.63\times10^{-34}\times6\times10^{14}}{1.6\times10^{-19}} = 2.48$ eV. $K_{max} = 2.48 - 2.14 = 0.34$ eV; $V_0 = 0.34$ V.

$v_{max} = \sqrt{\dfrac{2K}{m}} = \sqrt{\dfrac{2\times0.34\times1.6\times10^{-19}}{9.1\times10^{-31}}} \approx 3.5\times10^{5}$ m/s.

(Q11.5): slope $= \dfrac{h}{e} \Rightarrow h = 1.6\times10^{-19}\times4.12\times10^{-15} = 6.59\times10^{-34}$ J s.

Pattern C: de Broglie wavelength

Solved (Q11.10(a)): $\lambda = \dfrac{h}{mv} = \dfrac{6.63\times10^{-34}}{0.040\times1000} = 1.7\times10^{-35}$ m: far too small to observe, so wave nature is not seen for large objects.

(Q11.11): photon $p = \dfrac{h\nu}{c} \Rightarrow \lambda_{dB} = \dfrac{h}{p} = \dfrac{c}{\nu} = \lambda$.

Pattern A: Q11.1, 11.4. Pattern B: Q11.2, 11.3, 11.5, 11.6, 11.7, 11.8, 11.9. Pattern C: Q11.10, 11.11.

11D Conceptual questions

- **Effect of intensity on photocurrent and stopping potential?** → Current $\propto$ intensity; V_0 unchanged.
- **Effect of frequency?** → V_0 (and K_{max}) increases linearly; current unchanged (same intensity).
- **Why can't wave theory explain photoemission?** → It predicts no threshold frequency, K_{max} depending on intensity, and a time lag; none is observed.
- **Why is emission instantaneous?** → One photon gives its energy to one electron at once.
- **Same K: which has the shorter de Broglie λ, proton or electron?** → Proton ($\lambda \propto 1/\sqrt{m}$).
- **Same V: electron or α?** → α ($\lambda \propto 1/\sqrt{qm}$).

11E Exam tips

Common mistakes that lose marks

- Mixing eV and J: 1 eV $= 1.6 \times 10^{-19}$ J. Shortcut: $E(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})}$.
- K_{max} in eV equals V_0 in volts (same number).
- If $h\nu < \phi_0$: no emission at all, however intense the light.
- Use m_e for electrons, not m_p.

Question types: 1-mark: graphs (I–V, V_0–ν), threshold. 2–3 marks: Einstein's equation numericals, de Broglie wavelength. Case-based questions on photoelectric experiments are common.

11F JEE extras

Useful results

- Radiation pressure / force on a surface: $F = \dfrac{P}{c}$ (absorbed), $\dfrac{2P}{c}$ (reflected).
- Thermal neutrons: $\lambda = \dfrac{h}{\sqrt{3mkT}}$.
- Ratio of de Broglie wavelengths at same V: $\dfrac{\lambda_p}{\lambda_\alpha} = \sqrt{\dfrac{q_\alpha m_\alpha}{q_p m_p}} = 2\sqrt{2}$.

Example: electron through 100 V: $\lambda = \frac{1.227}{10} = 0.123$ nm.

11G Pattern spotter

If you see...	Do this	See
Photons		
Energy / momentum of a photon	$E = hc/\lambda$; $p = h/\lambda$	Ex A
Photons per second	$N = P/E$	Ex A
X-rays from electrons accelerated through V	$\nu_{max} = eV/h$	Ex A
Photoelectric effect		
ϕ_0 and ν given	$K_{max} = h\nu - \phi_0$; $V_0 = K_{max}/e$	Ex B
Will emission occur?	Compare $h\nu$ with ϕ_0	Ex B
Slope of V_0–ν graph	$h = e\times$ slope	Ex B
Matter waves		
Particle with m, v	$\lambda = h/mv$	Ex C
Electron through V volts	$\lambda = 1.227/\sqrt{V}$ nm	11A

Chapter 12 Atoms

12A Derivations (board favourites)

1. Distance of closest approach and impact parameter (Rutherford)

Head-on: all KE becomes PE at closest approach: $K = \dfrac{1}{4\pi\varepsilon_0}\dfrac{(2e)(Ze)}{d} \Rightarrow \boxed{d = \dfrac{2Ze^2}{4\pi\varepsilon_0 K}}$.

Small impact parameter $b \Rightarrow$ large scattering angle; most α-particles pass almost undeviated (atom mostly empty).

2. Bohr's model: radius, speed and energy

Postulates: stable orbits; $mvr = \dfrac{nh}{2\pi}$; photon $h\nu = E_i - E_f$. Centripetal force $= \dfrac{mv^2}{r} = \dfrac{e^2}{4\pi\varepsilon_0 r^2}$.

Solve with quantisation: $\boxed{r_n = \dfrac{n^2h^2\varepsilon_0}{\pi me^2} = n^2a_0}$ ($a_0 = 0.53$ Å), $v_n = \dfrac{e^2}{2\varepsilon_0 nh} \propto \dfrac{1}{n}$.

$K = \dfrac{e^2}{8\pi\varepsilon_0 r}$, $U = -\dfrac{e^2}{4\pi\varepsilon_0 r}$, $E = K + U = -\dfrac{e^2}{8\pi\varepsilon_0 r} \Rightarrow \boxed{E_n = -\dfrac{13.6}{n^2}\ \text{eV}}$. Note $K = -E$, $U = 2E$.

3. Spectral series (Rydberg formula)

$\dfrac{1}{\lambda} = R\left(\dfrac{1}{n_f^2} - \dfrac{1}{n_i^2}\right)$, $R = 1.097 \times 10^7\ \text{m}^{-1}$.

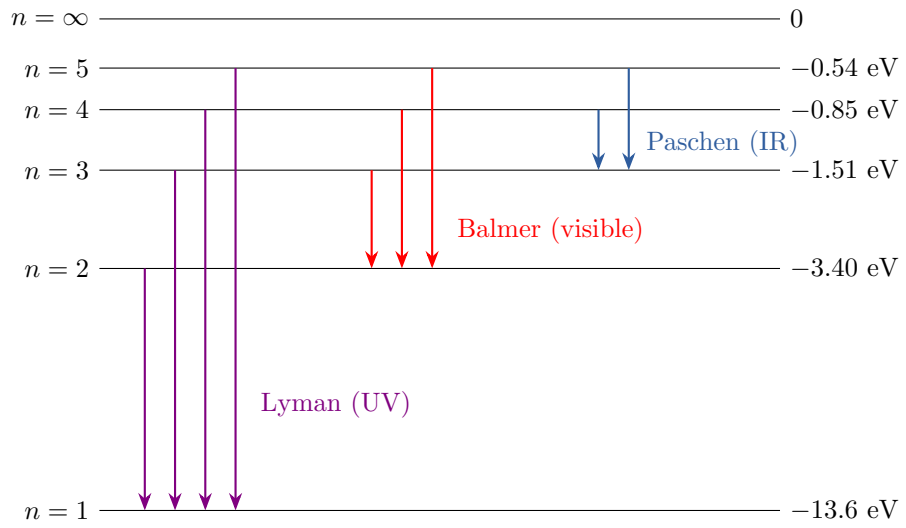
Lyman $n_f = 1$ (UV); Balmer $n_f = 2$ (visible); Paschen 3, Brackett 4, Pfund 5 (IR).

4. de Broglie's explanation of Bohr's postulate

Electron wave forms a standing wave round the orbit: $2\pi r_n = n\lambda = n\dfrac{h}{mv} \Rightarrow mvr = \dfrac{nh}{2\pi}$.

12B Diagrams to practise

Energy levels of hydrogen and the spectral series



Also practise: the α-scattering set-up (source, gold foil, rotatable detector) and the trajectory of an α-particle for different impact parameters.

12C Exercise-wise numerical patterns

Exercises 12.1–12.9 (9 questions, 3 patterns)

Pattern A: Atomic models (conceptual)

Solved (Q12.2): the α-particle (6.64×10^{-27} kg) is about 4 times heavier than the proton target, so it cannot bounce back even in a head-on collision: no large-angle scattering, unlike with gold.

Pattern B: Energy levels and transitions

Solved (Q12.5): $\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 12.75$ eV; $\lambda = \dfrac{1240}{12.75} = 97$ nm; $\nu = \dfrac{\Delta E}{h} = 3.1 \times 10^{15}$ Hz.
(Q12.8): a 12.5 eV electron can raise H to $n = 3$ (needs 12.09 eV) but not $n = 4$ (12.75 eV). Lines: $3 \to 1$ (103 nm) and $2 \to 1$ (122 nm) in Lyman; $3 \to 2$ (656 nm) in Balmer.
(Q12.4): $E = -13.6$ eV $\Rightarrow K = +13.6$ eV, $U = -27.2$ eV.

Pattern C: Bohr orbits (radius, speed, quantum number)

Solved (Q12.7): $r_n = n^2 r_1$: $r_2 = 4 \times 5.3 \times 10^{-11} = 2.12 \times 10^{-10}$ m, $r_3 = 9 \times 5.3 \times 10^{-11} = 4.77 \times 10^{-10}$ m.
(Q12.9): $n = \dfrac{2\pi mvr}{h} = \dfrac{2\pi \times 6 \times 10^{24} \times 3 \times 10^{4} \times 1.5 \times 10^{11}}{6.63 \times 10^{-34}} \approx 2.6 \times 10^{74}$: quantisation is unnoticeable for planets.

Pattern A: Q12.1, 12.2. Pattern B: Q12.3, 12.4, 12.5, 12.8. Pattern C: Q12.6, 12.7, 12.9.

12D Conceptual questions

- **Why did Rutherford's model fail?** → An accelerating (orbiting) electron should radiate and spiral into the nucleus; spectra would be continuous.
- **Why is the total energy negative?** → The electron is bound to the nucleus.
- **What does α-scattering show?** → Most of the atom is empty; positive charge and mass concentrated in a tiny nucleus.
- **Ionisation energy of H?** → 13.6 eV.
- **Limitation of Bohr's model?** → Only for one-electron atoms; can't explain intensities or fine structure.

12E Exam tips

Common mistakes that lose marks

- E_n is negative; energy of emitted photon = $E_{upper} - E_{lower}$ (positive).
- Shortest λ in a series: transition from $n = \infty$; longest: from the next level.
- $r \propto n^2$, $v \propto \frac{1}{n}$, $E \propto \frac{1}{n^2}$ (for H; multiply by Z, Z, Z^2 for H-like ions).
- Number of possible lines from level n: $\frac{n(n-1)}{2}$.

Question types: 1-mark: series in each region, ratio questions. 2–3 marks: energy/wavelength of transitions. 5 marks: Bohr's derivation of radius and energy, energy level diagram.

12F JEE extras

Useful results

- Hydrogen-like ion: $r_n = \dfrac{0.53n^2}{Z}$ Å, $E_n = -\dfrac{13.6Z^2}{n^2}$ eV, $v_n = \dfrac{2.18 \times 10^6 Z}{n}$ m/s.
- Time period $T \propto \dfrac{n^3}{Z^2}$; frequency of revolution $\propto \dfrac{Z^2}{n^3}$.
- Recoil of atom: $p = \dfrac{h\nu}{c}$, $v = \dfrac{h\nu}{mc}$.

Example: He^+ ground state: $E = -13.6 \times 4 = -54.4$ eV.

12G Pattern spotter

If you see...	Do this	See
Bohr model		
Energy of level n	$E_n = -13.6/n^2$ eV	Ex B
K and U from E	$K = -E$, $U = 2E$	Ex B
Photon for a transition	$h\nu = \Delta E$; $\lambda = 1240/\Delta E$ nm	Ex B
Lines emitted after excitation	Highest level reachable, then all downward jumps	Ex B
Radius / speed of orbit n	$r \propto n^2$, $v \propto 1/n$	Ex C
Quantum number of an orbit	$n = 2\pi mvr/h$	Ex C
Models		
Thomson vs Rutherford	Size same; mass distribution and stability differ	Ex A
Light target in α-scattering	No large-angle scattering	Ex A

Chapter 13 Nuclei

13A Derivations (board favourites)

1. Nuclear radius and density

$R = R_0 A^{1/3}$, $R_0 = 1.2$ fm. Density $\rho = \dfrac{mA}{\frac{4}{3}\pi R_0^3 A} = \dfrac{3m}{4\pi R_0^3}$: **independent of** A ($\approx 2.3 \times 10^{17}$ kg m^{-3}).

2. Mass defect and binding energy

$\Delta M = [Z m_p + (A - Z) m_n] - M_{nucleus}$ (or with m_H and atomic masses).

$\boxed{E_b = \Delta M\, c^2}$, 1 u $\times\, c^2 = 931.5$ MeV. Binding energy per nucleon $E_{bn} = \dfrac{E_b}{A}$.

3. Binding energy curve: fission and fusion

E_{bn} rises steeply for light nuclei, peaks near $A \approx 56$ (8.75 MeV), flat (≈ 8 MeV) for $30 < A < 170$, falls to 7.6 MeV for $A = 238$.

Fission: a heavy nucleus splits into two medium ones with higher E_{bn}: energy released (≈ 200 MeV per U-235).

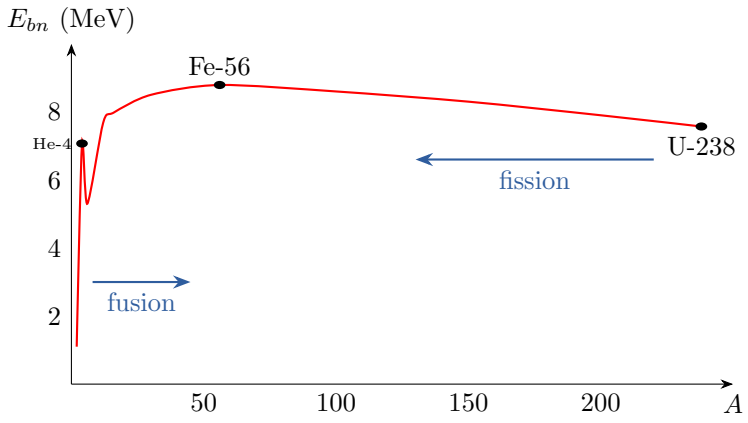
Fusion: light nuclei combine to a heavier one with higher E_{bn}: energy released (Sun, stars).

4. Q-value of a reaction

$Q = [\text{mass of reactants} - \text{mass of products}]c^2$. $Q > 0$: exothermic; $Q < 0$: endothermic.

13B Diagrams to practise

Binding energy per nucleon vs mass number (labels: peak near Fe-56, flat region, fission and fusion directions)



13C Exercise-wise numerical patterns

Exercises 13.1–13.10 (10 questions, 5 patterns)

Pattern A: Binding energy from masses

Solved (Q13.1): $\Delta M = 7m_H + 7m_n - m(^{14}\text{N}) = 7(1.007825) + 7(1.008665) - 14.00307 = 0.11236$ u.
$E_b = 0.11236 \times 931.5 = 104.7$ MeV. (For a coin, Q13.3: find E_b per atom, then multiply by the number of atoms.)

Pattern B: Nuclear size and density

Solved (Q13.4): $\dfrac{R_{Au}}{R_{Ag}} = \left(\dfrac{197}{107}\right)^{1/3} = 1.23$. **(Q13.10):** see derivation 1: ρ is independent of A.

Pattern C: Q-value of a reaction

Solved (Q13.5(i)): reactants: $1.007825+3.016049 = 4.023874$ u; products: $2\times2.014102 = 4.028204$ u.
$Q = (4.023874 - 4.028204) \times 931.5 = -4.03$ MeV: **endothermic**.
(Q13.6): $Q = [55.93494 - 2(27.98191)] \times 931.5 = -26.9$ MeV < 0: this fission is not possible.

Pattern D: Energy from fission / fusion of a given mass

Solved (Q13.7): atoms in 1 kg of Pu-239: $N = \dfrac{1000}{239} \times 6.023 \times 10^{23} = 2.52 \times 10^{24}$.
Energy $= 2.52 \times 10^{24} \times 180 = 4.5 \times 10^{26}$ MeV.
(Q13.8): 2.0 kg of deuterium: reactions $= \frac{1}{2} \times \frac{2000}{2.014} \times 6.023 \times 10^{23}$; energy $\times 3.27$ MeV; time $= \frac{E}{100\ \text{W}} \approx 4.9 \times 10^4$ years.

Pattern E: Coulomb barrier

Solved (Q13.9): deuterons just touch at $r = 2R = 4.0$ fm: $U = \dfrac{ke^2}{r} = \dfrac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{4 \times 10^{-15}} = 5.76 \times 10^{-14}$ J $= 360$ keV.

Pattern A: Q13.1, 13.2, 13.3. Pattern B: Q13.4, 13.10. Pattern C: Q13.5, 13.6. Pattern D: Q13.7, 13.8. Pattern E: Q13.9.

13D Conceptual questions

- **Properties of nuclear force?** → Strongest force, short range ($\sim$ few fm), charge independent, attractive (repulsive below ≈ 0.8 fm), saturates.
- **Why is E_{bn} nearly constant for $30 < A < 170$?** → Saturation: each nucleon interacts only with its near neighbours.
- **Why is fusion hard on Earth?** → Very high temperature needed to overcome the Coulomb barrier.
- **Isotopes / isobars / isotones?** → Same Z / same A / same N.
- **Why is the mass of a nucleus less than the sum of its nucleons?** → The mass defect appears as binding energy.
- **Three types of radioactive decay?** → α (He nucleus), $\beta^{\pm}$ (electron/positron), γ (photon).

13E Exam tips

Common mistakes that lose marks

- Use m_H (not m_p) with *atomic* masses so the electron masses cancel.
- 1 u $= 931.5$ MeV$/c^2$; 1 MeV $= 1.6 \times 10^{-13}$ J.
- Sign of Q: reactants minus products.
- Number of atoms: $\frac{\text{mass (g)}}{\text{molar mass}} \times N_A$.

Question types: 1-mark: nuclear force properties, isotopes, density independence. 2–3 marks: binding energy, Q-value. 5 marks: BE/nucleon curve with fission and fusion explanation.

13F JEE extras

Useful results

- Decay law: $N = N_0 e^{-\lambda t}$; half-life $T_{1/2} = \dfrac{0.693}{\lambda}$; mean life $\tau = \dfrac{1}{\lambda}$; activity $A = \lambda N$; after n half-lives $N = N_0/2^n$.
- α decay: $A \to A-4$, $Z \to Z-2$; β^-: $Z \to Z+1$; β^+: $Z \to Z-1$.
- Energy shared in α-decay: $K_\alpha = Q\dfrac{A-4}{A}$.

Example: after 3 half-lives, $\frac{1}{8}$ of the sample remains.

13G Pattern spotter

If you see...	Do this	See
Masses and energy		
Binding energy of a nucleus	$\Delta M \times 931.5$ MeV	Ex A
Energy to separate all nucleons in a sample	E_b per atom $\times$ number of atoms	Ex A
Ratio of radii	$(A_1/A_2)^{1/3}$	Ex B
Reactions		
Is a reaction exo/endothermic?	Sign of Q	Ex C
Is fission possible?	$Q > 0$?	Ex C
Energy from 1 kg of fuel	Number of nuclei $\times$ energy per event	Ex D
Lamp/power running time	$t = E/P$	Ex D
Barrier height	ke^2/(distance between centres)	Ex E

Chapter 14 Semiconductor Electronics

14A Key explanations (board favourites)

1. Energy bands: conductors, semiconductors, insulators

Conductor: bands overlap or CB partly filled. Insulator: $E_g > 3$ eV. Semiconductor: $E_g < 3$ eV (C 5.4 eV, Si 1.1 eV, Ge 0.7 eV).
Intrinsic: $n_e = n_h = n_i$. Extrinsic: $n_e n_h = n_i^2$.

2. Doping

n-type: pentavalent donor (P, As, Sb); electrons majority, holes minority.
p-type: trivalent acceptor (B, Al, In); holes majority, electrons minority. Both remain electrically neutral.

3. Formation of a p-n junction

Holes diffuse from p to n and electrons from n to p (concentration gradient). They leave behind immobile ions: the **depletion region**. The resulting field drives a **drift** current opposite to diffusion; at equilibrium they are equal and a **barrier potential** (≈ 0.7 V for Si) exists.

4. Diode under bias

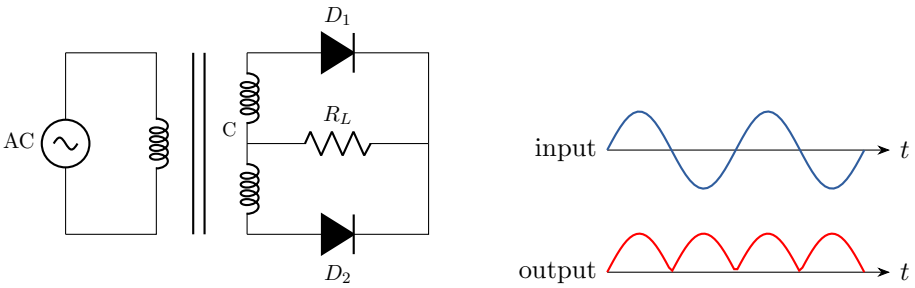
Forward (p to +): barrier lowered, depletion layer narrows, current (mA) rises sharply above the threshold voltage.
Reverse (p to −): barrier raised, depletion layer widens, tiny reverse current (μA) until breakdown.

5. Rectifiers

Half-wave: one diode conducts in alternate half cycles; output frequency = input frequency.
Full-wave (centre-tap): two diodes conduct alternately; current through the load in the same direction both halves; output frequency = 2× input. A capacitor filter smooths the output.

14B Diagrams to practise

Full-wave rectifier (centre-tap) with input and output waveforms



Also practise: energy band diagrams (n-type with donor level just below CB, p-type with acceptor level just above VB); p-n junction showing the depletion region and barrier; V–I characteristics of a diode (forward and reverse).

14C Exercise-wise solution patterns

Exercises 14.1–14.6 (6 questions, 3 patterns)

Pattern A: Doping, carriers and band gaps

Solved (Q14.1, 14.2): n-type: electrons majority, holes minority, pentavalent dopant: **(c)**. p-type: holes majority, trivalent dopant: **(d)**.
(Q14.3): $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$: **(c)**.

Pattern B: p-n junction behaviour

Solved (Q14.4, 14.5): holes diffuse because their concentration is higher in the p-region: **(c)**. Forward bias lowers the potential barrier: **(c)**.

Pattern C: Rectifier output

Solved (Q14.6): half-wave: output ripple at 50 Hz (one pulse per cycle); full-wave: 100 Hz (two pulses per cycle).

Pattern A: Q14.1, 14.2, 14.3. Pattern B: Q14.4, 14.5. Pattern C: Q14.6.

14D Conceptual questions

- **Effect of temperature on a semiconductor's resistance?** → Decreases: more electron–hole pairs.
- **Is a doped semiconductor charged?** → No, it stays electrically neutral.
- **Why is the depletion region free of mobile carriers?** → Diffusing carriers recombine, leaving only fixed ions.
- **Why is reverse current almost independent of voltage (before breakdown)?** → It is due to minority carriers, limited by their number.
- **Which is better for a rectifier, Si or Ge?** → Si: lower reverse current, works at higher temperature.
- **Width of depletion layer in forward/reverse bias?** → Decreases / increases.

14E Exam tips

Common mistakes that lose marks

- Forward bias: p-side to the *positive* terminal.
- In p-type, electrons are minority carriers (not absent).
- Diffusion current: majority carriers; drift current: minority carriers.
- Label circuit diagrams fully: diodes, centre tap, load, input and output waveforms.

Question types: 1-mark: majority carriers, bias effects, band gap order, output frequency. 3 marks: p-n junction formation, V–I characteristics. 5 marks: full-wave rectifier with circuit and waveforms.

14F JEE extras

Useful results

- Conductivity: $\sigma = e(n_e\mu_e + n_h\mu_h)$.
- Diode current: $I = I_0(e^{eV/kT} - 1)$.
- Logic gates: AND ($Y = A \cdot B$), OR ($A + B$), NOT ($\bar{A}$); NAND and NOR are universal gates.
- Zener diode: used as a voltage regulator in reverse breakdown.

Example: $n_i = 1.5 \times 10^{16}$ m^{-3}, doped to $n_e = 4.5 \times 10^{22}$ m^{-3}: $n_h = \frac{n_i^2}{n_e} = 5 \times 10^{9}$ m^{-3}.

14G Pattern spotter

If you see...	Do this	See
Materials		
Majority carriers / dopant	n: electrons, pentavalent; p: holes, trivalent	Ex A
Order of band gaps	C > Si > Ge	Ex A
Minority carrier density	$n_e n_h = n_i^2$	14A
Junction and rectifiers		
Why diffusion across junction	Concentration difference	Ex B
Forward / reverse bias	Barrier lowered / raised	Ex B
Output frequency	HW: f; FW: $2f$	Ex C